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Original Article

Sexual and reproductive health experiences and care of adult women with cystic fibrosis

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ABSTRACT

Background: As survival and health improve in people with cystic fibrosis (CF), more women with CF (wwCF) are considering their sexual and reproductive health (SRH). This study compared SRH experiences, behaviors, and care utilization of wwCF to the general population and defined CF-impacted considerations and care preferences.
Methods: We surveyed wwCF aged ≥25 years regarding SRH and compared results to the US National Survey of Family Growth (NSFG;n = 4357) and friend controls(n = 123). We used descriptive statistics and chi-squared/Fisher's exact testing and linear regression for comparisons.
Results: A total of 460 wwCF (mean age 36.1 years) completed the survey. WwCF were less likely to report current contraceptive use (43%vs76% NSFG, p<0.001;60% friends, p = 0.005). Nearly 25% of wwCF reported worsened CF symptoms during their menstrual cycles, 50% experienced urinary incontinence, and 80% vulvovaginal candidiasis. WwCF were significantly less likely to be parents (46%vs62% friends, p = 0.015) and to have experienced pregnancy (37%vs78% NSFG, p<0.001;58% friends, p = 0.002). More wwCF required medical assistance to conceive (29%vs12% NSFG, p<0.001 and 5% friends, p<0.001). Eighty-four percent of wwCF view their CF doctor as their main physician and 41% report no primary care provider (vs19% friends; p<0.001). WwCF report suboptimal rates of contraceptive and preconception counseling/care and are less likely to have received HPV vaccination (42%vs55%friends, p = 0.02). Despite desiring SRH conversations with their CF team, <50% report discussing SRH topics.
Conclusion: WwCF have significantly different SRH experiences than non-CF peers. They report suboptimal SRH care compared to their preferences highlighting an urgent need to encourage SRH counseling/care in the CF model.
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1. Introduction

Half of people with cystic fibrosis (CF) born from 2016 to 2020 in the United States (US) are predicted to live beyond 50 years of age [1]. This is largely due to the development of CF transmembrane conductance regulator (CFTR) modulators that correct

Abbreviations: SRH, sexual and reproductive health; wwCF, women with cystic fibrosis; HEMT, highly effective modulator therapy; CFTR, CF transmembrane conductance regulator; AYA, adolescent and young adult; STI, sexually transmitted infection.

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the basic defect in CF [2]. Based on CFTR variants, approximately 90% of people with CF ≥6 years are eligible for highly effective modulator therapy (HEMT), including ivacaftor and elexacaftor-tezacaftor-ivacaftor [3,4]. With CFTR modulators, more women with CF (wwCF) are reassessing their reproductive goals and desire sexual and reproductive health (SRH) discussions with their CF care team [5,6]. In fact, the approval and use of HEMT in the US coincide with a rise in pregnancies in CF [7]. As wwCF increasingly face both general and disease-specific considerations, a better understanding of SRH experiences and care throughout the lifespan is needed [8].

Our prior work demonstrates that adolescent and young adult(AYA) wwCF face significant SRH concerns and gaps in care provision [9,10]. Despite similar rates of sexual activity, wwCF ages 15–24 years were less likely to have ever used contraception or been tested for sexually transmitted infections (STIs) in the past year compared to the general US population [10]. Furthermore, a third reported delayed puberty, one-sixth urinary incontinence (UI), nearly half vulvovaginal candidiasis, and one-sixth sexual dysfunction [9].

This study describes the general and disease-impacted SRH experiences and care of wwCF ≥25 years of age and compares these findings to those of peers and the general population. We hypothesized that wwCF would have similar SRH behaviors, but lower SRH care utilization rates, compared to healthy counterparts. Furthermore, we define rates of SRH concerns related to CF-specific impacts and care preferences to inform improvements to SRH CF care.

2. Methods

From January 2019-September 2020, we recruited wwCF ≥25 years of age at 10 US CF centers to participate in an anonymous survey investigating SRH topics including sexual history, STI history, contraceptive use/counseling, pregnancy/parenthood experiences and attitudes, and receipt of SRH services, We also examined SRH considerations related to CF pulmonary and genitourinary symptoms as they relate to SRH and preferences for SRH discussions and care.

Our prior qualitative and survey-based work guided survey content [5,6,9,10]. We derived select survey items from the US National Survey of Family Growth (NSFG), the Female Sexual Functioning Index-6 (FSFI-6) and validated intimate partner violence measures [11–13]. NSFG is a nationally representative SRH survey of US people age 15–49 years ($n = 6141$ in 2017–2019 sample) [11]. We further refined the survey via cognitive interviews with two wwCF using think-aloud and verbal probing techniques [14].

Research coordinators (RCs) at each site recruited patient participants in-person to complete the survey during a regularly scheduled clinic visit or during hospitalization. They separately tracked enrollment of participants to assure that wwCF were not approached more than once to complete the survey. RCs asked patient participants to nominate one female friend without CF to complete a similar SRH survey to create a demographically similar control group. RCs shared the survey link with nominated friends if they consented to participate. To encourage the disclosure of SRH behaviors and experiences, we collected data anonymously using REDCap (Research Electronic Data Capture) [15]. The Institutional Review Board of each site approved this study; the University of Pittsburgh served as the coordinating site.

We utilized descriptive statistics to summarize respondent demographics as well as general and CF-impacted SRH considerations. We compared demographics and general SRH experiences and behaviors for wwCF in our sample ages 25–49 years to data for 25–49-year-old women from the NSFG and compared our entire CF sample to friends. We represented continuous variables using means and standard errors for the CF and friend samples and using weighted mean and standard errors for the NSFG; similarly, we represented categorical variables with percentages endorsed for the CF and friend samples, and weighted percentages endorsed for the NSFG. We excluded missing values from denominators in percent calculations. We further explored differences between wwCF, the NSFG, and friends via chi-squared and Fisher's exact testing (for cell counts ≤ 10) for categorical variables and linear regression for continuous variables. For testing between CF and NSFG, we added Rao-Scott corrections to our chi-squared testing. In our comparisons, we used NSFG's standard sampling weights and structure to account for the survey design, including considerations related to base probability of selection weights, nonrespondents, and post-stratification adjustments for age, sex, and race/ethnicity [11]. To assess differences in SRH across CFTR modulator use, we compared wwCF taking HEMT, moderately effective modulators, and those taking no modulators. We conducted statistical analyses using SAS version 9.4 (SAS Institute, Cary, NC, USA; 2016) and SAS Enterprise Guide version 8.2 (SAS Institute, Cary, NC, USA; 2019).

3. Results

3.1. Demographic and CF-related health characteristics

A total of 460 wwCF (mean age 36.1 years) completed the survey out of 1148 eligible participants across all centers (40% response rate). We compared data among 25–49-year-old wwCF to 25–49-year-old women from the NSFG ($n = 4357$; mean age 36.7 years). These groups had significant differences for age, marital status, race/ethnicity, level of education, and working status. We compared the entire sample of wwCF to the friend sample ($n = 123$; mean age 33.4 years) with noted significant differences related to age and working status (Table 1).

Forty-three percent of wwCF rated their CF as "mild", 50% as "moderate", and 7% as "severe". Forty-nine percent self-reported a baseline forced expiratory volume in 1 second (FEV1) of >70% predicted; 57% had been hospitalized in the past year. Twenty percent were taking HEMT at the time of the survey; however, it should be noted that approval for elexacaftor-tezacaftor-ivacaftor did not occur until October 2019, after most respondents had already completed the survey (Supplemental Table 1).

3.2. Sexual activity, STIs, and contraception use

Over 96% of all groups reported ever having vaginal sex with a male. WwCF were significantly older at first vaginal sex with a male compared to NSFG and reported significantly more lifetime male sexual partners and partners in the prior year compared to NSFG. There were no significant differences between wwCF and friends related to sexual activities (Table 2).

Nearly two-thirds of wwCF (64%) always disclose their CF diagnosis to sexual partners. Factors that determined CF disclosure included being in a defined/committed relationship (25%), comfort (24%), and relationship length (18%).

Eight percent of wwCF reported ever being diagnosed with an STI compared to 33% in NSFG ($p<0.001$). There was no significant difference in STI diagnosis between wwCF and friends. WwCF were significantly less likely to report undergoing STI testing in the past year compared to NSFG (25% vs. 32%) with no differences from friends (Table 2). Six percent of wwCF report receiving a STI diagnosis/treatment in the past year compared to 4% of NSFG ($p = 0.001$), but there were no differences with friends. WwCF were significantly less likely to report a history of pelvic inflammatory disease compared to the NSFG (3% vs.5%) with no differences compared to friends.

Nearly all wwCF, NSFG women, and friends reported ever using contraception. WwCF age 25–49-years were significantly less likely

Table 1
Sample Demographics for Women with CF, NSFG, and Friend Controls.

Demographic	CF % (n) Ages 25–49 (n = 401)	NSFG Weighted% Ages 25–49 (n = 4357)	p-value	CF % (n) (n = 460)	Friend Control % (n) (n = 123)	p-value
Age in years (mean, SE)	33.3 (0.3)	36.7 (0.2)	<0.001	36.1 (0.5)	33.4 (0.8)	0.017
Marital status	n = 400		<0.001	n = 459	n = 110	0.758
Formerly married	5.4 (22)	11.5		7.4 (34)	5.5 (6)	
Living with partner	12.8 (52)	15.3		12.4 (57)	10 (11)	
Married	52.5 (210)	51.5		53.6 (246)	55.5 (61)	
Single	29 (116)	21.7		26.6 (122)	29.1 (32)	
Race/ethnicity	n = 399		<0.001	n = 457	n = 110	0.536
Hispanic	4.3 (17)	19.6		3.9 (18)	6.4 (7)	
Non-Hispanic Other	4 (16)	23.3		3.9 (18)	3.6 (4)	
Non-Hispanic White	91.7 (366)	57.0		92.1 (421)	90 (99)	
Level of education	n = 399		<0.001	n = 458	n = 110	0.056
College degree or higher	66.9 (267)	38.6		66.4 (304)	78.2 (86)	
High school degree/GED or less	10 (40)	32.6		10.5 (48)	6.4 (7)	
Some college or vocational school	23.1 (92)	28.9		23.1 (106)	15.5 (17)	
Working status	n = 401		<0.001	n = 460	n = 109	<0.001
Attending school	5.7 (23)	3.0		5 (23)	1.8 (2)	
Homemaker	12.2 (49)	15.8		11.7 (54)	17.4 (19)	
Not working/Other	3.7 (15)	6.2		5.7 (26)	3.7 (4)	
"Not working/in school because of health"	16.7 (67)	N/A		19.1 (88)	2.8 (3)	
Working full or part time	61.6 (247)	75.0		58.5 (269)	74.3 (81)	
Sexual orientation	n = 400		.096	n = 458	n = 104	0.519
Bisexual	7.3 (29)	7.1		6.6 (30)	9.6 (10)	
Heterosexual or straight	91.3 (365)	90.9		91.9 (421)	88.5 (92)	
Homosexual, gay, or lesbian	1.5 (6)	2.0		1.5 (7)	1.9 (2)	

to report currently using contraception than NSFG (43% vs.76%) and compared to friends (43% vs.60%).

Table 2 summarizes contraceptive use/choice for all samples. WwCF were significantly less likely to report using female or male sterilization as their current contraceptive method compared to NSFG (4% vs.40% and 4% vs.14%). Aside from sterilization, wwCF were significantly more likely to report use of all contraceptive methods compared to NSFG. No differences in current contraceptive choice were noted compared to friends. WwCF reported similar reasons for other uses of contraception compared to friends except for control of endometriosis (12% vs.21%, p = 0.02).

WwCF were significantly less likely to report consistent condom use with male partners compared to NSFG. More wwCF reported using emergency contraception compared to NSFG (35% vs.23%, p<0.001). There were no significant differences between wwCF and friends (Table 2).

3.3. General and CF-impacted srh experiences and behaviors

Sixty-five percent of wwCF reported having regular menstrual periods. Among those with irregular periods, 39% reported that they were using contraception that impacted menstruation, and 7% reported that they reached or were approaching menopause. WwCF were significantly more likely to report a hysterectomy compared to friends (15% vs.3%). Nearly one-quarter of wwCF reported that their CF symptoms worsened during their menstrual cycle (24%). Two-thirds reported that symptoms worsen immediately prior to, and 27% concurrently with, menstruation. One quarter reported using contraception to help control CF symptoms during menstruation (Supplemental Table 2).

WwCF were significantly more likely to experience urinary incontinence (UI) than friends (50% vs.31%). Thirty percent of wwCF reported experiencing UI once a week or more (with 14% reporting daily episodes). WwCF and friends had a different profile of UI triggers (Supplemental Table 2). Forty-five percent of wwCF with UI reported that it sometimes or often inhibited completion of chest physiotherapy.

WwCF were significantly more likely to report vulvovaginal candidiasis than friends (80% vs.56%) with 27% reporting occurrence ≥2/year. One-third of wwCF reported a lifetime history of intimate partner violence and 7% reproductive coercion (i.e., partner exerted pressure to become pregnant and/or engaged in contraceptive sabotage) with no significant differences compared to friends. Twenty-four percent of wwCF who had sexual experiences in the past month had a composite score on the FSFI-6 signaling sexual dysfunction with no significant differences compared to friends. Forty-five percent of wwCF reported coughing during or after sex ≥50% of the time; 5% reported hemoptysis, and 9% excessive flatulence. WwCF did not experience ovarian cysts, fibroid tumors/myomas, endometriosis, or problems with ovulation/menstruation at rates significantly different from friends (Supplemental Table 2).

Supplemental Table 3 highlights rates of general and CF-impacted SRH outcomes among wwCF by use of HEMT, moderately effective, or no CFTR modulator. Overall, there were no significant differences between groups except the reported UI trigger of sneezing.

3.4. Parenthood and pregnancy

Nearly half of 25–49-year-old wwCF (46%) desired future children compared to 40% of NSFG and 53% of friends (p<0.001 and p = 0.146). Forty-six percent of 25–49-year-old wwCF were concerned that future children might have CF. Nine percent reported being told that they were infertile because of CF. WwCF were significantly more likely to report being non-parents compared to friends (54% vs.38%). Regarding parenthood paths, wwCF were significantly more likely to pursue adoption and less likely to achieve

Table 2
Sexual activity and contraception use.

Sexual Activity	Women with CF% (n) Ages 25–49 $n = 401$	NSFG Weighted% Ages 25–49 $n = 4357$	p value	Women with CF% (n) Ages 25+ $n = 453$	Friend Control % (n) Ages 25+ $n = 88$	p value
Ever had vaginal sex with male partner	96.0 (381)	97.4	<0.001	96.4 (429)	98.9 (87)	0.330
Age at first sex[a], mean (SE)	18 (0.2)	17.3 (0.1)	<0.001	18.1 (0.2)	18.9 (0.3)	0.103
Ever performed oral sex on male partner	93.7 (371)	84.9	<0.001	93.5 (414)	96.6 (84)	0.333
Ever performed oral sex on female partner	11.7 (47)	12.7	0.139	11.4 (50)	11.4 (10)	1
Ever received oral sex from male partner	94.4 (373)	86	<0.001	94.1 (417)	95.4 (83)	0.802
Ever received oral sex from female partner	12.2 (60)	14.8	0.003	12 (53)	13.6 (12)	0.722
Ever had anal sex	38.6 (151)	40.4	0.205	38 (167)	39.8 (35)	0.810
Ever had any sexual experience with a female	15.2 (60)	20.2	<0.001	14.5 (64)	17 (15)	0.515
Number of male sexual partners, lifetime[a], mean (SE)	11.0 (1.0)	7.6 (0.2)	<0.001	10.8 (1)	10.1 (1.8)	0.519
Number of male sexual partners[a], past 12 months, mean (SE)	1.1 (0.0)	1.0 (0.0)	<0.001	1.1 (0.8)	1.5 (0.3)	0.004
Sexually transmitted infections						
Lifetime history of STI diagnosis (gonorrhea, chlamydia, herpes, HIV, syphilis, and/or genital warts)	8.4 (32)	33.3	<0.001	7.9 (33)	7.3 (6)	1
Lifteime history of HPV diagnosis				41.2 (178)	36.4 (32)	0.694
Receipt of any STI testing in past 12 months	24.7 (99)	32.2	<0.001	22.2 (102)	22.8 (28)	0.888
Receipt of any STI diagnosis or treatment in past 12 months	5.5 (22)	3.8	0.001	5.2 (24)	3.3 (4)	0.480
History of pelvic inflammatory disease (PID) treatment	3.1 (12)	4.7	<0.001	3.9 (17)	4 (4)	1
Contraception						
Any lifetime BC use	94.9 (370)	97.3	<0.001	92.9 (405)	96.6 (85)	0.242
Any current BC use[b]	43.1 (173)	75.6	<0.001	43.1 (173)	60.2 (50)	0.005
Current method used, among current BC user[c]						
Female sterilization	4 (7)	40.4	<0.001	4.6 (8)	6 (3)	0.712
Male sterilization	3.5 (6)	14.3	<0.001	3.4 (6)	4 (2)	1
IUD	37.6 (65)	15.5	<0.001	37.7 (66)	38 (19)	1
Implant	2.9 (5)	2	0.009	2.9 (5)	2 (1)	1
Injectable	7.5 (13)	2.1	<0.001	7.4 (13)	4 (2)	0.531
Pill	27.7 (48)	18	<0.001	28 (49)	40 (20)	0.119
Patch	0.6 (1)	0.3	0.074	0.6 (1)	0 (0)	1
Vaginal ring	4.6 (8)	1.2	<0.001	4.6 (8)	0 (0)	0.204
Diaphragm	0 (0)	0	–	0 (0)	0 (0)	–
Male condoms	24.9 (43)	16.6	<0.001	24.6 (43)	18 (9)	0.447
Withdrawal	10.4 (18)	4.4	<0.001	10.3 (18)	8 (4)	0.790
Other uses of contraception						
Cramps or pain during				52.3 (228)	44 (37)	0.190
periods				23.6 (103)	27.4 (23)	0.487
Acne				11.5 (50)	21.4 (18)	0.020
Endometriosis				5 (22)	3.6 (3)	0.782
Regulate menstrual periods						
Condom use frequency, past 12 months[d]						
Every time	11.1 (38)	15.4	<0.001	10.3 (38)	9 (7)	0.830
Most of the time	6.2 (21)	5.5		6 (22)	6.4 (5)	
Some to half of the time	11.7 (40)	10.5		10.9 (40)	14.1 (11)	
None of the time	71 (242)	68.6		72.8 (268)	70.5 (55)	
Lifetime emergency contraception use[a]	35.1 (131)	23	<0.001	31.8 (133)	37.9 (33)	0.316

[a] Of those who ever had vaginal sex with a male, $n = 381$ women with CF, $n = 4236$ Unweighted NSFG denominator, $n = 102$ friend control denominators.
[b] Of women under age 49 (limiting CF and friend control groups), $n = 401$ women with CF, $n = 84$ friend control denominators.
[c] Of current contraceptive users, $n = 173$ women with CF, $n = 2895$ Unweighted NSFG, and $n = 52$ friend control denominators.
[d] Of those who had sex in the past year, $n = 348$ women with CF, $n = 3600$ Unweighted NSFG denominator, $n = 92$ friend control denominators.

Table 3
Parenthood and pregnancy.

	Women with CF% (n) Ages 25–49 $n = 401$	NSFG Weighted% Ages 25–49 $n = 4357$	P value	Women with CF% (n) $n = 460$	Control % (n) $n = 123$	P value
Parenthood Status						
No children				54.3 (250)	38.2 (47)	0.002
Yes, through pregnancy				32 (147)	42.3 (52)	0.032
Yes, through adoption	4.2 (17)	1.3	<0.001	5.2 (24)	0.8 (1)	0.041
Yes, through surrogacy/gestational carrier				1.3 (6)	0 (0)	0.351
Yes, through step-parenthood				3 (14)	4.1 (5)	0.570
Yes, through foster parenting				0.7 (3)	0 (0)	1
Pregnancy History						
Ever pregnant	37.4 (140)	77.8	<0.001	40.3 (170)	57.7 (56)	0.002
Number of pregnancies[a] mean(se)	1.9 (0.09)	2.9 (0.05)	<0.001	1.9 (0.08)	2.2 (0.15)	0.049
Any live births[a]	80 (112)	91.9	<0.001	82.4 (140)	89.3 (50)	0.293
Number of live births[a] mean(se)	1.0 (0.07)	2.0 (0.04)	<0.001	1.3 (0.06)	1.7 (0.13)	0.293
Ever had an abortion[a]	12.1 (17)	17.6	<0.001	15.3 (26)	8.9 (5)	0.270
Ever had a miscarriage				25.9 (44)	30.4 (17)	0.513
Ever had a stillbirth				0 (0)	0 (0)	
Required medical assistance to conceive[a]	28.8 (40)	11.7	<0.001	29 (49)	5.4 (3)	<0.001
Preconception and Prenatal Care						
Received preconception counseling[a]				49.4 (84)	30.4 (17)	0.013
Lack of prenatal care[b]				7.2 (10)	4 (2)	0.736
Parenthood/Pregnancy Attitudes						
Desire to have a future child [c]						
Definitely no	36.6 (140)	59.0	<0.001	36.6 (140)	25.3 (21)	0.146
Definitely yes	26.2 (100)	39.2		26.2 (100)	25.3 (21)	
Probably no	17.8 (68)	0.9		17.8 (68)	21.7 (18)	
Probably yes	19.4 (74)	1.0		19.4 (74)	27.7 (23)	
Timing if pregnancy happened today [c]						
At an ok, but not quite right time				27.1 (103)	30.1 (25)	0.066
At the right time				20 (76)	14.5 (12)	
At the wrong time				25.5 (97)	25.3 (21)	
Too late				18.9 (72)	12 (10)	
Too soon				8.4 (32)	18.1 (15)	
Feelings regarding prospect of pregnancy at this time[c]						
A little pleased				18.2 (69)	15.7 (13)	0.025
A little upset				17.7 (67)	33.7 (28)	
I would not care				4.7 (18)	2.4 (2)	
Very pleased				27.4 (104)	18.1 (15)	
Very upset				31.9 (121)	30.1 (25)	
Breastfeeding of prior children[d]						
Yes				47.8 (66)	60 (30)	0.158
Combination of bottle/breast feeding				23.2 (32)	24 (12)	

[a] Of those who ever been pregnant.
[b] Among those who had ever had a live birth.
[c] Of those age less than or equal to 49 years.
[d] Of those who have had children.

parenthood through pregnancy compared to friends (5% vs.1% and 32% vs.42%) (Table 3).

Thirty-seven percent of 25–49-year-old wwCF reported ever being pregnant compared to 78% of NSFG ($p<0.001$). Forty percent of wwCF reported prior pregnancy compared to 58% of friends ($p = 0.002$). WwCF reported significantly fewer lifetime pregnancies compared to NSFG and friends. Pregnant wwCF were also significantly less likely to have experienced a live birth or abortion compared to pregnant NSFG women. There was no significant difference in pregnancy outcomes compared to friends. Twenty-nine percent of wwCF required medical assistance to conceive compared to 12% of NSFG and 5% of friends ($p<0.001$ and $p<0.001$).

WwCF who reported a pregnancy were significantly more likely to have received preconception counseling compared to friends

(49% vs.30%). Among wwCF who underwent a live birth, 7% reported no prenatal care compared to 4% of friends ($p = 0.3$). Forty-eight percent of wwCF who experienced prior live births reported breastfeeding their children, and 23% reported a combination of breast and formula feeding, similar to friends (Table 3).

Supplemental Table 4 highlights pregnancy/parenthood outcomes by use of CFTR modulator. WwCF taking HEMT were significantly more likely to be parents, achieve parenthood via pregnancy, and been pregnant compared to those taking moderately effective or no modulators. There were no significant differences related to pregnancy outcomes, desires, or breastfeeding.

3.5. Sexual and reproductive health care utilization

Eighty-four percent of wwCF view their CF physician as their "main doctor". Fifty-nine percent identify a primary care provider (PCP) compared to 81% of friends ($p<0.001$). WwCF reported significantly fewer PCP visits than friends, with 27% reporting seeing their PCP "very rarely" compared to 9% of friends (Table 4).

WwCF were significantly less likely to report ever receiving emergency contraception compared to friends (9% vs.17%). Similar rates exist between both groups regarding other SRH services. When considering SRH services received in the past year, wwCF were significantly more likely to report receiving contraceptive and emergency contraception counseling, emergency contraception provision, and pregnancy testing than NSFG. No significant differences exist between the groups in relation to receiving a Pap smear/pelvic exam in the past year. When compared with friends, wwCF did not show any significant differences in accessing these services in the past year (Table 4).

Thirty-eight percent of wwCF reported receiving any SRH service from their CF team. Approximately one quarter (24%) reported receiving a method or prescription for contraception and 30% contraceptive counseling from their CF team. Eight percent obtained STI testing at their CF center and 1% received STI counseling. One third took a pregnancy test at their CF center. Forty-seven percent reported receiving counseling about family planning or pregnancy.

Forty-two percent of 25–49-year-old wwCF reported receiving at least one HPV vaccine compared to 55% of similarly-aged friends ($p = 0.02$). Four percent had received an HPV vaccine at their CF center. Among those who reported receiving ≥1 shot, two-thirds of wwCF completed the full HPV vaccination series compared to 71% of friends ($p = 0.6$) (Table 4).

Nearly all wwCF and friends reported undergoing a Pap smear; there were no significant differences regarding the age of first Pap smear or the timing of their last Pap smear (Table 4). Nine percent of wwCF were clearly nonadherent with the timing of their last Pap smear based on current guidelines compared to 6% of friends ($p = 0.4$). Nineteen percent of wwCF reported a history of an abnormal Pap smear in the prior 3 years compared to 25% of friends ($p = 0.2$). Fifty-nine percent of wwCF reported undergoing a prior HPV test compared to 53% of friends ($p = 0.3$). WwCF and friends did not exhibit any differences regarding their perception of the need/timing of their next Pap smear and HPV testing (Table 4).

Approximately one-third of wwCF reported undergoing a mammogram compared to 29% of friends ($p = 0.2$). Both wwCF and their friends had similar reasons for receiving a mammogram. Most wwCF and friends aged ≥40 years had undergone a mammogram (88% vs.82%, $p = 0.6$) (Table 4).

3.6. SRH discussions with cf providers

Table 5 summarizes SRH discussions with the CF team. Fifty-five percent of wwCF reported that they had ever discussed menstrual periods with their CF provider with 19% of those respondents reporting that this occurred once a year or more. Approximately half reported ever discussing pregnancy and contraception with the CF team (52% and 49%, respectively). Forty-four percent reported ever discussing vulvovaginal candidiasis, 41% fertility, 34% healthy relationships, and 34% parenthood. The minority of respondents who reported SRH discussions had such conversations at least annually. Most respondents reported that SRH discussions occurred with their CF physician or advanced practice provider.

3.7. SRH care preferences

Fig. 1 summarizes the preferences of wwCF regarding SRH. Two-thirds selected at least one SRH topic that they would like to discuss with their CF team. Ninety-one percent would like their CF physician or advanced practice provider to have SRH discussions. Fifty-six percent prefer to have the CF team initiate SRH discussions with 52% wanting no partners/family in the room. Fifty-two percent of wwCF are interested in seeing a SRH specialist as part of their CF care visit.

4. Discussion

This study is the largest to date investigating the general and disease-impacted SRH experiences and care of wwCF compared to people without CF. We identified several unique symptoms and patterns that warrant further exploration.

This study employed two comparison groups: the general US population as represented in the NSFG and a friend control group. As the currently-diagnosed CF population is demographically different than the general US population (particularly related to race/ethnicity), the comparisons with demographically-similar friends may be informative of differences driven by CF status. For instance, compared to NSFG, wwCF reported significantly different use rates of most types of contraception (most strikingly female/male sterilization) and were significantly less likely to use condoms, have a history of or testing for STIs, and have used emergency contraception. However, comparisons with friends demonstrated no significant differences in these areas. Notably, wwCF were significantly less likely to report current contraceptive use compared to the NSFG *and* friends, suggesting that the difference may be driven by aspects of CF.

Nearly 25% of wwCF in our sample reported that CF symptoms worsen during their menstrual cycle. This is unsurprising as it is theorized that female sex hormones play a role in sex-based disparities in CF outcomes, including overall survival, pulmonary exacerbations, and microbiologic profile [16–19]. A recent prospective study evaluating the effects of female sex hormones and hormonal contraception on lung function and inflammation found a significant increase in sputum proinflammatory markers and a corresponding decrease in lung function during ovulation (the time of peak estrogen) [20]. Further research is needed on the impact of sex hormones on respiratory health in CF and ways to optimize management [21].

Comparable with AYA wwCF [9], wwCF aged ≥25 years engage in similar sexual behaviors as their non-CF peers but are less likely to use contraception and to experience pregnancy. As expected, due to accumulated effects of disease over time, more adult wwCF reported experiencing UI, vulvovaginal candidiasis, and sexual dysfunction than 15–24-year-old wwCF [9]. Our prior work among younger wwCF had notably limited pregnancy/parenthood experiences; this sample provides key data related to impaired fertility, historically decreased pregnancy rates, and alternative paths to parenthood. We identified that wwCF are significantly less likely to be parents and more likely to require medical assistance to conceive than the general population or non-CF peers.

Encouragingly, this sample demonstrated better SRH care utilization than AYA wwCF when compared to peers without CF

Table 4
Sexual and reproductive health care utilization.

	Women with CF% (n) Ages 25–49 *n* = 401	NSFG Weighted% Ages 25–49 *n* = 4357	p value	Women with CF% (n) Ages 25+ *n* = 453	Friend Control % (n) Ages 25+ *n* = 88	p value
View CF physician as "main doctor"				84.3 (387)		
Identifies a primary care provider (PCP)				59 (271)	80.9 (89)	<0.001
Frequency of PCP visits						
Once every 3 months or more often				28.9 (130)	5.6 (5)	<0.001
Once every 6 months				17.1 (77)	16.9 (15)	
Once a year				20 (90)	43.8 (39)	
Once every 1–2 years				7.3 (33)	24.7 (22)	
Very rarely				26.7 (120)	9 (8)	
General SRH care utilization						
Services received at clinic or doctors' office EVER						
Contraceptive counseling				32.5 (140)	37.5 (36)	0.346
Emergency contraception				9.3 (40)	16.5 (16)	0.045
Emergency contraception counseling				3 (13)	7.4 (7)	0.069
Pregnancy test				43.8 (189)	52 (51)	0.137
Postnatal care[a]				89.1 (123)	95.8 (46)	0.246
STI testing or treatment				22.5 (96)	27.4 (26)	0.314
Pap smear or pelvic exam				90.9 (399)	96 (96)	0.106
Services received at clinic or doctors' office PAST 12 months[b]						
Contraceptive counseling	20.8 (25)	13.9	<0.001	19.6 (27)	28.6 (10)	0.246
Emergency contraception	8.3 (3)	1.5	<0.001	7.9 (3)	0 (0)	0.547
Emergency contraception counseling	9.1 (1)	1.2	<0.001	8.3 (1)	0 (0)	1
Pregnancy test	25.3 (40)	18.2	<0.001	21.9 (40)	26 (13)	0.536
Pap smear or pelvic exam	61.8 (214)	63.4	.300	60 (234)	52.2 (48)	0.171
HPV vaccination						
Receipt of HPV vaccine (at least 1 shot)[c]				41.5 (164)	54.7 (52)	0.020
Receipt of the complete HPV vaccine series (3 shots)[c,d]				67.1 (110)	71.2 (37)	0.582
Cervical cancer screening						
Ever had a Pap smear				96.6 (423)	97 (97)	1
Age at first Pap smear in years (mean, SE)				24 (4.8)	19.6 (0.4)	0.661
Timing of last Pap smear[e]						
A year ago or less				53.9 (228)	56.7 (55)	0.810
Between 1 and 5 years ago				41.4 (175)	40.2 (39)	
Over 5 years ago				4.7 (20)	3.1 (3)	
Timing of last Pap smear clearly nonadherent with current guidelines[f]				9.1 (39)	5.6 (5)	0.403
History of abnormal Pap smear in last 3 years				18.5 (69)	25 (22)	0.168
Ever had HPV test				59.4 (183)	52.9 (37)	0.315
Age at first HPV test in years (mean, SE)				23.1 (0.5)	22.2 (1)	0.484
Perceived timing of next Pap smear for regular cancer screening				40.2 (174)	37 (37)	0.099
Every year				22.2 (96)	20 (20)	
Every 2 years				19.9 (86)	32 (32)	
Every 3 years				0.9 (4)	0 (0)	
Every 4 years				10.2 (44)	9 (9)	
Every 5 years				6.7 (29)	2 (2)	
Less often than every 5 years						

(*continued on next page*)

Table 4 (*continued*)

	Women with CF% (n) Ages 25–49 $n = 401$	NSFG Weighted% Ages 25–49 $n = 4357$	p value	Women with CF% (n) Ages 25+ $n = 453$	Friend Control % (n) Ages 25+ $n = 88$	p value
Perceived timing of next HPV						
test for regular cancer				24 (100)	23.1 (21)	0.713
screening				15.4 (64)	16.5 (15)	
Every year				13.2 (55)	15.4 (14)	
Every 2 years				1.9 (8)	1.1 (1)	
Every 3 years				13.5 (56)	7.7 (7)	
Every 4 years				32 (133)	36.3 (33)	
Every 5 years						
Less often than every 5 years						
Mammograms						
History of mammogram				36.5 (160)	29 (29)	0.155
Age at first mammogram in years (mean, SE)				33.6 (0.7)	31.7 (1.1)	0.245
Main reason for first						0.1
mammogram				53.5 (85)	34.5 (10)	0.107
Routine				33.3 (53)	41.4 (12)	
Lump/problem				10.7 (17)	24.1 (7)	
Family history of cancer				2.5 (4)	0 (0)	
Other						
Mammogram complete among those age 40 years or older (% yes)				88.1 (104)	81.8 (9)	0.627

[a] Among those who had ever had a live birth, $n = 140$ women with CF, $n = 50$ friend controls.
[b] Among those who ever received SRH services.
[c] In 2006, the HPV vaccine was approved to age 26 years and was expanded in 2018 to include all females age 27–45 years. Thus, we truncated our sample in include respondents ages ≤49 years old as they would be the only ones eligible for vaccination.
[d] Among those who reported receiving at least 1 HPV vaccine shot.
[e] Among those who ever received a Pap smear.
[f] The 2020 guidelines for Pap smear timing is the following: age 21–29 years should have a Pap test every 3 years; age 30–65 years should have a Pap test and an HPV test (co-testing) every 5 years, have a Pap test alone every 3 years, or have an HPV test alone every 5 years. We defined non-adherence with guidelines in this dataset as respondents age 21–29 years who go longer than 3 years without undergoing a Pap test and respondents age 30–65 years who go longer than 5 years without a Pap test.

Table 5
SRH Discussions with CF Providers.

	Frequency of discussions with CF Provider		CF Provider who discussed topic [a,b]			
	% (n) $n = 460$		% (n)			
SRH topic	Ever	Once a year or more[c]	CF doctor/advanced practice provider[d]	CF social worker	CF nurse	Other CF provider[e]
Menstrual periods	55.2 (254)	18.9 (48)	92.1 (232)	6 (15)	22.6 (57)	15.5 (39)
Urinary incontinence	24.3 (112)	8 (9)	73 (81)	2.7 (3)	12.6 (14)	31.5 (35)
Vulvovaginal candidiasis	43.9 (202)	8.9 (18)	89.6 (180)	0.5 (1)	31.8 (64)	2.5 (5)
Healthy relationships	34.1 (157)	17.2 (27)	58.8 (90)	62.1 (95)	15 (23)	19 (29)
Sexual orientation	7 (32)	0 (0)	69.7 (23)	33.3 (11)	18.2 (6)	15.2 (5)
Sexual activity	29.8 (137)	8 (11)	85.9 (116)	14.8 (20)	17.8 (24)	7.4 (10)
Sexual functioning	7.6 (35)	2.9 (1)	80 (28)	11.4 (4)	22.9 (8)	20 (7)
STIs	10 (46)	0 (0)	84.8 (39)	17.4 (8)	21.7 (10)	4.3 (2)
Contraception	48.5 (223)	17 (38)	94.5 (206)	7.8 (17)	21.6 (47)	5 (11)
Fertility	41.3 (190)	7.4 (14)	97.4 (187)	9.9 (19)	12 (23)	5.7 (11)
Pregnancy	52 (239)	12.6 (30)	95.8 (228)	16.4 (39)	19.3 (46)	16.4 (39)
Breastfeeding	17 (78)	5.1 (4)	89.6 (69)	5.2 (4)	19.5 (15)	33.8 (26)
Parenthood	33.7 (155)	27.7 (43)	93.5 (145)	38.7 (60)	21.3 (33)	16.1 (25)
Talking to partner about CF and SRH	24.8 (114)	5.3 (6)	87.4 (97)	19.8 (22)	13.5 (15)	7.2 (8)
Menopause	8 (37)	10.8 (4)	94.6 (35)	2.7 (1)	16.2 (6)	8.1 (3)
Cervical cancer screening/Pap smear	13 (60)	8.3 (5)	88.1 (52)	6.8 (4)	15.3 (9)	6.8 (4)
Breast cancer screening	10.4 (48)	12.5 (6)	93.8 (45)	6.3 (3)	18.8 (9)	0 (0)

[a] Among those who responded they had SRH discussions with any CF provider choice.
[b] Respondents can select multiple provider types.
[c] Among those who report ever having a SRH discussion with their CF provider on this topic.
[d] Including physician assistants or nurse practitiioners.
[e] Other CF provider includes nutritionists, respiratory therapists, physical therapists, and psychologists.

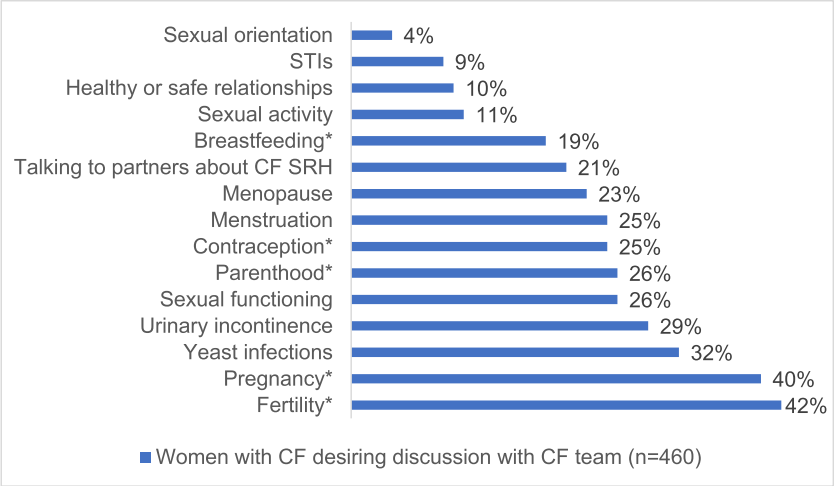

Fig. 1. Patient Preferences Regarding SRH Discussions
*Of those aged ≤45 years, $n = 377$.

[10]. These findings support prior work highlighting that adult CF providers are superior at SRH care provision than pediatric colleagues [22]. Additionally, adult wwCF may have increased self-efficacy in seeking SRH care. However, the rates of SRH care utilization overall remain suboptimal with <1/3rd of wwCF reporting that they received contraceptive and emergency contraceptive counseling or STI testing and <50% of those previously pregnant receiving preconception counseling. WwCF are significantly less likely to report HPV vaccination than non-CF peers. This is particularly concerning as HPV is the principal risk factor for cervical cancer and both transplanted and non-transplanted wwCF experience high rates of abnormal Pap smears and cervical dysplasia [23,24].

Our results on the perceived role of CF providers as "de facto" PCPs align with prior research [25]. In 2020, a US CF center conducted a chart review to uncover gaps in primary care provision among adults with CF [26]. In 59% of abstracted charts, there was no PCP identified or the primary CF provider was listed as the PCP; overall, there was poor documentation of recommended CF-unrelated screening. Qualitative researchers highlight the need for a PCP or SRH specialist to have a familiarity with modern CF care to assure successful care provision [5,27].

An important consideration when interpreting our findings is the timing of this survey and the potential SRH impacts of widespread HEMT use. Only 20% of wwCF in our study sample were taking HEMT, compared to the approximately 90% who are currently eligible. The number of reported pregnancies among US wwCF nearly doubled in the year following the approval of the HEMT elexacaftor-tezacaftor-ivacaftor, from 310 in 2019 to 619 in 2020 [1]. The cause of this increase remains unclear and is likely multifactorial, with HEMT potentially improving fertility through changes in cervical mucus and vaginal pH. Better lung function and quality of life are also influencing reproductive goals [28]. WwCF taking HEMT in our study were significantly more likely to be parents, parents through pregnancy, and to have ever been pregnant compared to those taking moderately effective or no CFTR modulator. The long-term impact of HEMT on UI, vulvovaginal candidiasis, and sexual functioning has not been explored; however, comparisons in our sample by short-term modulator use and type did not identify major differences.

Our survey findings continue to highlight a discrepancy between wwCF's preferences and experiences related to SRH discussions/care with the CF team. WwCF desire routine, tailored SRH education and discussions initiated by their CF team beginning in adolescence [6]. Ongoing work attempts to address these needs via patient-facing education and decision support tools [29–31]. Moreover, robust provider training curricula and evaluation tools to improve SRH care for people with CF is needed [32]. We also should further explore how to directly integrate SRH within CF care through novel systems-based interventions [33,34]. Use of relationship-centered communication skills, such as those in the CF Foundation's Partnership Enhancement Program, can be applied to SRH discussions with wwCF and may help CF teams discuss SRH [35]. Specifically, adaptation and implementation of frameworks such as the Power to Decide's "One Key Question" may empower CF providers to routinely ask wwCF about family-building desires and goals and offer personalized counseling and care based on their response [36]. While such frameworks often focus on pregnancy, a broader focus on becoming a parent (through any means) may be more applicable for wwCF. As each CF center and team has unique practice environments and skills, quality improvement initiatives around optimizing SRH should be individualized.

We recognize this study's limitations. As an anonymous, self-report survey design, this study did not verify CF and SRH clinical data. Participants' recall bias may impact reports of SRH experiences and care. In addition, despite the anonymous nature of the study, eligible participants may not have wanted to participate in the study for various reasons; the characteristics and responses of non-respondents may therefore be significantly different from respondents to the survey. Aside from gathering feedback on the instrument's face validity, we did not conduct additional psychometric assessment. Importantly, participants who identify as black, indigenous, and people of color are underrepresented in our sample compared to the general CF population. We also did not include items related to several important factors that may influence SRH care and decisions (such as insurance status, religious/personal preferences, and genotype). Finally, and importantly, this work was limited to the SRH experiences of cis-gender females with CF; future work should explore the SRH experiences and preferences of gender diverse CF populations.

In conclusion, research around SRH in CF should continue to be prioritized, especially in the age of widespread childhood initiation of HEMT [37]. There is an urgent need to improve access to SRH care and SRH care delivery by encouraging patient-provider SRH conversations with the CF team with wwCF of all ages. Innovative partnerships with specialists and PCPs and quality improvement

initiatives may benefit SRH care and may improve outcomes for wwCF. Importantly, CF teams should encourage all wwCF to have a PCP for both general and applicable SRH-specific concerns.

Credit author statement

Traci M. Kazmerski was responsible for: Conceptualization; Funding acquisition; Cystic Fibrosis Foundation; Investigation; Methodology; Project administration; Resources; Software; Validation; Roles/Writing - original draft; Writing - review & editing. Olivia Stransky was responsible for: Data curation; Methodology; Project administration; Writing - review & editing. Daniel Lavage and Alana Fields were responsible for: Data curation; Formal analysis; Validation; Writing - review & editing. Jennifer Taylor-Cousar, Gregory Sawicki, Sigrid Ladores, Emily M. Godfrey, Moira Aitken, Sandra Sufian, Manu Jain, Tara Barto, Joanne Billings, and Denis Hadjiliadis were responsible for: Investigation; Writing – review & editing. Raksha Jain was responsible for: Conceptualization; Funding acquisition; Investigation; Methodology; Project administration; Supervision; Visualization; Writing - review & editing.

Declaration of Competing Interest

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jcf.2022.09.013.

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